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Hypothesis Testing Problem set

1. To test the hypothesis that a coin is fair, the following decision rule is adopted:

i) Accept the hypothesis if the number of heads in a single sample of 100 tosses is between 40 and 60 inclusive.

ii) Reject the hypothesis otherwise.

a) find the probability of rejecting the hypothesis when it is actually correct.

b) interpret graphically the decision rule and the result of part (a).

c) what conclusion would you draw if the sample of 100 tosses yielded 53 heads? 60 heads?

d) Could you be wrong in your conclusion to (c)? Explain.

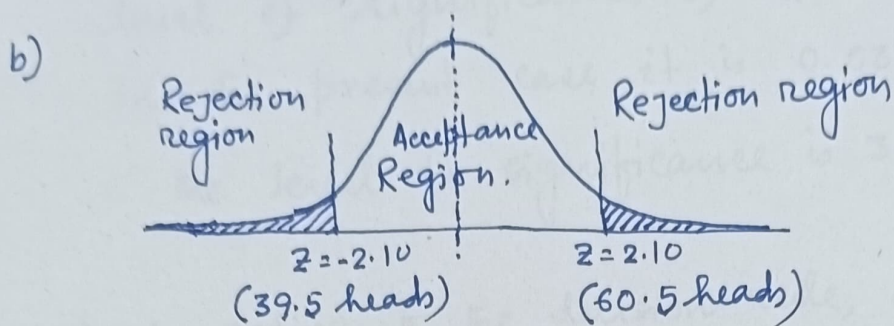
Ans:

a) In the previous problem we calculated the probability of getting 40 and 60 heads inclusive and found = 0.9642.

∴ The probability of not getting between 40 and 60 heads inclusive if the coin is fair equals $1 - 0.9642 = 0.0358$.

* Thus this is the probability that the coin is fair but the number of heads in a single toss sample of 100 toss is ~~not~~ not between 40-60 inclusive.

∴ The probability of rejecting the hypothesis when it is correct equals 0.0358.



The decision rule is illustrated in the above figure. It shows the probability distribution of heads in 100 tosses of a fair coin.

If a single sample of 100 tosses yields a z-score between -2.10 and +2.10, we accept the hypothesis; otherwise we reject the hypothesis and decide that the coin is not fair.

The error made in rejecting the hypothesis when it should be accepted (False Negative) is the Type I error of the decision rule. and the probability of making this error, equal to 0.0358 is represented by the shaded area of the figure.

If

Level of significance

If a single sample of 100 tosses yields a number of heads whose z score lies in the shaded regions, we would say that this z-score differed significantly from what it should be expected if the hypothesis were true.

For this reason the total shaded area (i.e. probability of a type I error) is called the level of significance of the decision rule. In the present case it is $0.0358 = 3.58\%$.

$\therefore$ the level of significance is 3.58% .

- c) According to the decision rule, we would have to accept the hypothesis that the coin is fair in both cases.

One might argue that if only one more head had been obtained, we would have rejected the hypothesis. This is what one must face when any sharp line of decision is used in making decisions.

- d) Yes. We could accept the hypothesis when it actually should be rejected, as would be the case, for example, when the probability of heads is really 0.7 instead of 0.5 .

The error made in accepting the hypothesis when it should be rejected (False positive) is the type II ~~data~~ error of the decision.

Ex: 2 Design a decision rule to test the hypothesis that a coin is fair if a sample of 64 ~~heads~~ tosses of the coin is taken and if a level of significance (a) 0.05 (b) 0.01 is used.

a) If the level of significance is 0.05 then each shaded area is 0.025 by symmetry.

Then the area between 0 to z_1 is —

$$0.5 - 0.025 = 0.4750.$$

from the table it corresponds — $z_1 = 1.96$.

Thus a possible decision rule is:

(1) Accept the hypothesis that the coin is fair if z is between -1.96 and 1.96 .

(2) Reject the hypothesis otherwise.

To express this decision rule in terms of the number of heads to be obtained in 64 tosses of the coin we calculate the mean

$$\mu = np = 64 \times \frac{1}{2} = 32$$

$$\text{and } \sigma = \sqrt{npq} = \sqrt{64(0.5)(0.5)}$$

$$= 8 \times 0.5 = 4.$$

under the hypothesis that the coin is fair.

$$\text{Then } z = \frac{(X - \mu)}{\sigma} = \frac{(X - 32)}{4}$$

$$\text{If } z = 1.96 \Rightarrow \frac{(X - 32)}{4} = 1.96$$

$$\Rightarrow X = 39.84$$

$$\text{If } z = -1.96, \frac{(X - 32)}{4} = -1.96 \text{ or } X = 24.16.$$

Therefore, the decision rule becomes:

1) Accept the hypothesis that the coin is fair if the number of heads is between 24.16 and 39.84. i.e. between 25 and 39 inclusive.

2) Reject the hypothesis otherwise.

b) if the level of significance is 0.01 then the area of each shaded region = 0.005 by symmetry.

Then the area between 0 and z_1 is -

$$0.5 - 0.005 = 0.4950.$$

from the table it corresponds to $z = 0.575$
 ≈ 0.58 .

$$\therefore \frac{X - 32}{4} = 0.58 \Rightarrow X - 32 = \cancel{2.32} 10.32$$

$$\Rightarrow X = 42.32 \approx 42$$

and for $z = -2.58$

$$X - 32 = -10.32$$

$$\Rightarrow X \approx 22$$

$\therefore$ The decision rule becomes:

1) Accept the hypothesis if the number of head is between 22 and 42 inclusive.

2) Reject the hypothesis otherwise.